

1. Title:

- a. Early-Stage Drug Discovery Utilizing Deep Learning

2. Team Members:

- a. Stephen Marriott
- b. Will Welsh
- c. Wanming He

3. Introduction

- a. Our group is reimplementing and hopefully improving on the models outlined in the following paper:
- b. *DEEPScreen: high performance drug–target interaction prediction with convolutional neural networks using 2-D structural compound representations*
- c. <https://pubs.rsc.org/en/content/articlelanding/2020/sc/c9sc03414e>
- d. The goal behind this model is to predict active compounds in target proteins using a convolutional neural network.
- e. This is an area of promise in deep learning moving forward, and touches on the intersection of chemistry and health, which all team members involved found interesting.
- f. This is structured in the form of a binary classification problem, where the model will determine whether the compound is active or inactive for that specific target.

4. Related Work

- a. A number of papers and models use deep learning for drug discovery related tasks. One of the other models cited in the paper the team is working on implementing is *Visualizing convolutional neural network protein-ligand scoring*. In this paper, the researchers also implemented a CNN to predict the binding affinity of a protein-ligand complex. However, this paper utilized three-dimensional molecular data, something that our team is not planning on working with at this stage. This paper also focused on visualizing their results for interpretability, which can also help with analyzing the results and applying them to future drug discovery research.

5. Data

- a. Data for this project was obtained from the DEEPScreen GitHub repository, available here: <https://github.com/cansyl/DEEPScreen/blob/pytorch/README.md>
- b. For each target protein, the researchers behind the original paper created folders with images of the structure of component compounds, as well as a json file containing labels for whether the protein was active or not, and split the data into training, validation, and test sets.
- c. As each protein is unique, our plan is to train models for a smaller subset of the 800+ available target proteins.

6. Methodology

- a. Our first step has been to reimplement the model, which was initially trained using Pytorch, in TensorFlow.
- b. The original model relied on 5 convolutional layers, as well as 3 dense layers for the binary classification problem

- c. Our initial next step will be attempting to create a more accurate model by changing some of the hyperparameters and shapes for a model with similar architecture.
- d. Assuming these steps are successful, we think a reasonable next step will be adding some form of attention, either to a model trained on the underlying SMILES strings if they are available or the actual images themselves.

7. Metrics

- a. Accuracy is a reasonable metric to assess success for this project, as the goal is binary classification, although the original paper looked at precision and recall as well, so we may include those metrics in our final presentation.
- b. According to the paper, the model achieved greater than 80% accuracy on 613/704 target models, with an average accuracy of 87%.
- c. Our base level goal would be to replicate this accuracy in Tensorflow, with our target goal to make incremental improvements on the proteins we test by updating in smaller ways the hyperparameters. A stretch goal would be to achieve similar frequency of accuracy above 90%, assuming that attention in this context leads to some improvement.

8. Ethics

- a. Broadly, effective models for drug discovery are important to society because they guide effective clinical research, and suggest areas in which resources can be invested to lead to better disease treatment. While deep learning can be an effective approach to initial discovery, at this stage it is not a substitute for actual clinical research as part of the drug discovery process, and confidence in its results should not be overstated.
- b. Relatedly, accuracy is important, in particular avoiding false positive results, because inaccurate predictions could lead to wasted resources in advancing drug discovery and medical research. It is important to avoid misusing these scarce resources, so decisions guided by these models should be relatively certain in their predictions.

9. Division of labor

- a. At this stage, all three of us are working on updating the model design and contributing to GitHub.
- b. Stephen has more subject knowledge in this field, so he will be able to better provide background on the project for the poster
- c. Wanming and Will will focus on visualizing the model and results for the presentation, while helping in other areas when needed.